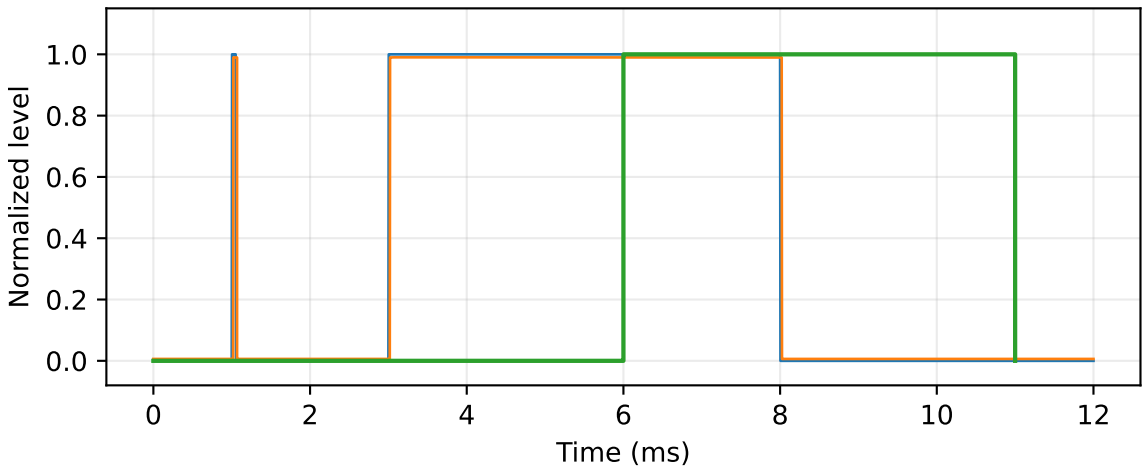
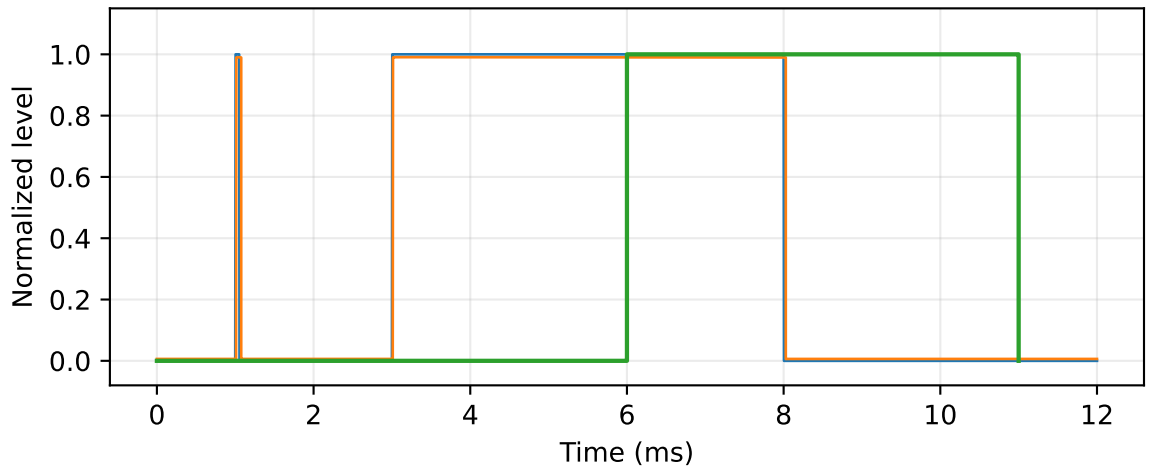
Input filtering and 2 ms firmware debounce — simulated, not bench measured

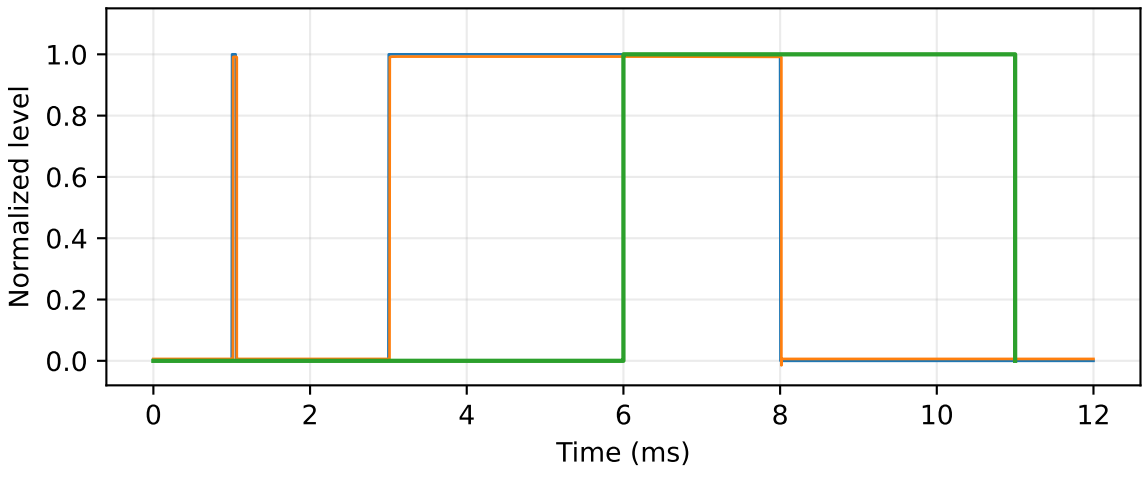
logic 3v3



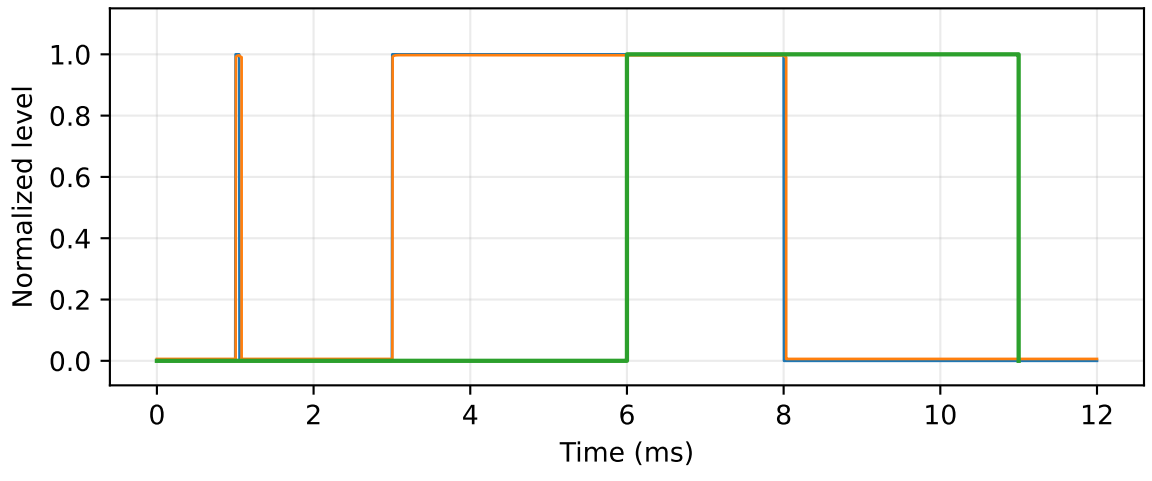
logic 5v



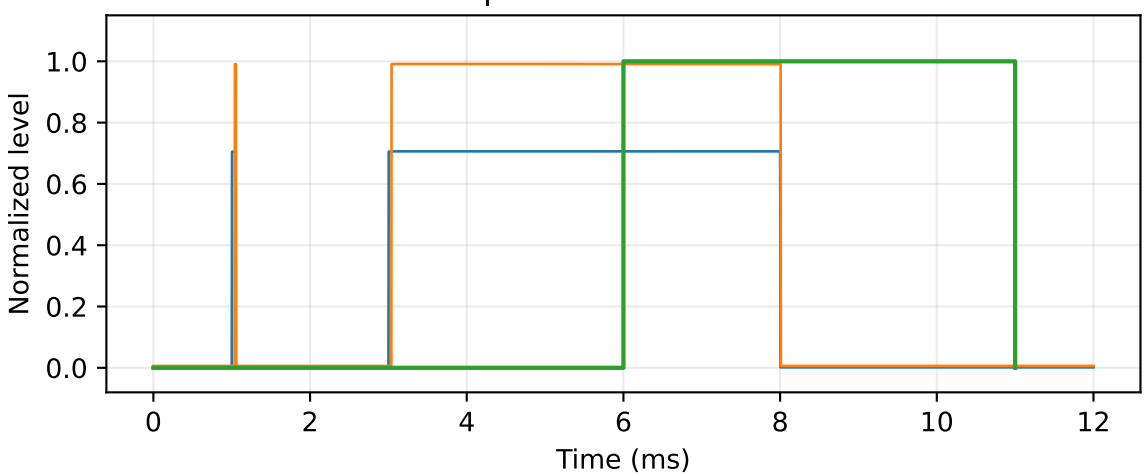
industrial 12v



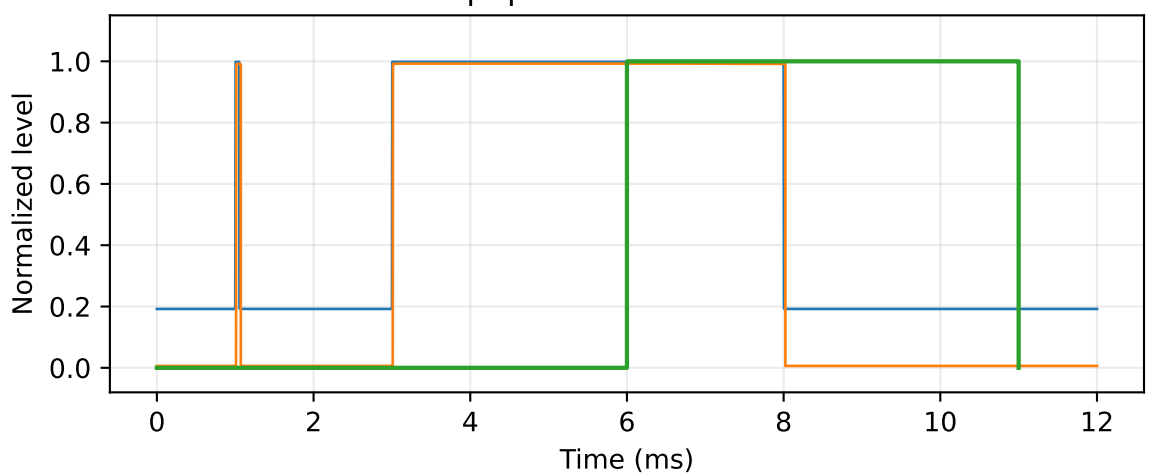
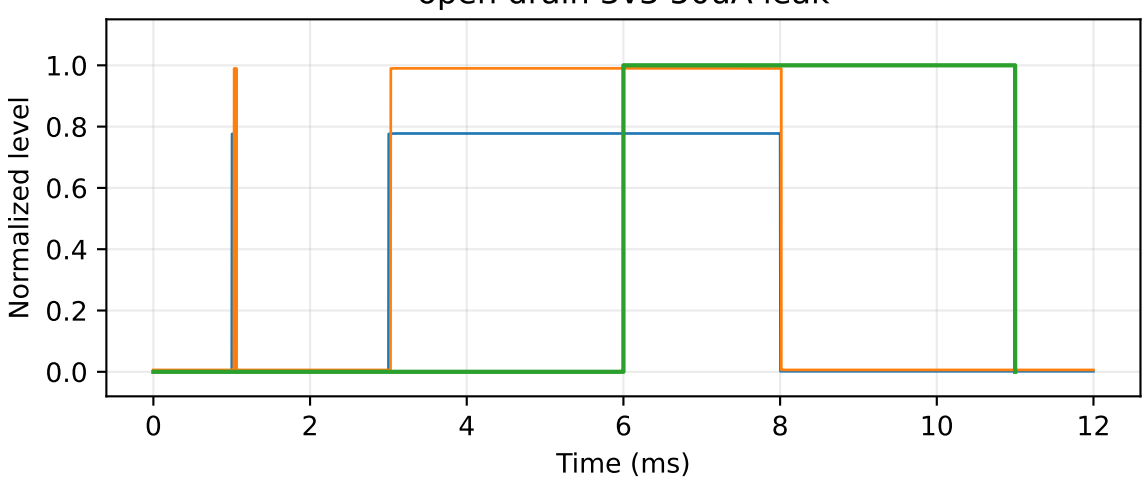
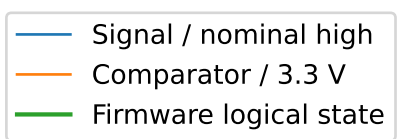
industrial 30v at 6v battery



npn 10v 500uA leak



pnp 12v 500uA leak



open drain 3v3 50uA leak